THE EXTENDED ALTERNATING CECH COMPLEX IS CONTRACTIBLE WHEN ONE OF THE ELEMENTS IS A UNIT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot, namely sources/stacks-chow/preamble.tex lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros). Nothing was added, removed or redefined, no package was added, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

Editorial note. The citation fields (source_id, source_version, authors, source_url, license.name, source_title) are identical to those of the cards already published from this chapter, so that the batch selector treats them as one source instead of deferring them as a same-source clash.

Allowed standard prerequisites (editorial).

- Extended alternating Cech complex of a sequence and of a module, and the shape of its differential. constructed in the discussion of the same section, source lines 6908-6943, immediately before the counted statement. Supplies the complex, the indexing of its cochains and the sign convention of its differential that the explicit computation uses.
- Localisation at f_i is an isomorphism when f_i is a unit. used at proof lines 7009-7016 to identify M of a multi-index with M of the same multi-index with f_i inserted. Makes the homotopy formula well defined by placing the new entry i into the multi-index at the correct position.
- A contraction of a complex exhibits it as homotopy equivalent to zero. used at proof lines 7017-7018, the announced strategy of the computation. Turns the explicit verification dh + hd = -id into the vanishing conclusion of the statement.

Source reference. Original lemma 0G6J: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20199–20199. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Lemma 0.2. *Let R be a ring. Let $f_1, \ldots, f_r \in R$. Let M be an R-module. If there exists an $i \in \{1, \ldots, r\}$ such that f_i is a unit, then the extended alternating Čech complex of M is homotopy equivalent to* 0.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. We will use the following notation: a cochain x of degree $p+1$ in the extended alternating Čech complex of M is $x = (x_{i_0 \ldots i_p})$ where $x_{i_0 \ldots i_p}$ is in $M_{f_{i_0} \ldots f_{i_p}}$. With this notation we have

$$d(x)_{i_0 \ldots i_{p+1}} = \sum\nolimits_j (-1)^j x_{i_0 \ldots \hat{i}_j \ldots i_{p+1}}$$

As homotopy we use the maps

$$h : \text{cochains of degree } p+2 \to \text{cochains of degree } p+1$$

given by the rule

$$h(x)_{i_0 \ldots i_p} = 0 \text{ if } i \in \{i_0, \ldots, i_p\} \text{ and } h(x)_{i_0 \ldots i_p} = (-1)^j x_{i_0 \ldots i_j i i_{j+1} \ldots i_p} \text{ if not}$$

Here j is the unique index such that $i_j < i < i_{j+1}$ in the second case; also, since f_i is a unit we have the equality

$$M_{f_{i_0} \ldots f_{i_p}} = M_{f_{i_0} \ldots f_{i_j} f_i f_{i_{j+1}} \ldots f_{i_p}}$$

which we can use to make sense of thinking of $(-1)^j x_{i_0 \ldots i_j i i_{j+1} \ldots i_p}$ as an element of $M_{f_{i_0} \ldots f_{i_p}}$. We will show by a computation that $dh + hd$ equals the negative of the identity map which finishes the proof. To do this fix x a cochain of degree $p+1$ and let $1 \leq i_0 < \ldots < i_p \leq r$.

Case I: $i \in \{i_0, \ldots, i_p\}$. Say $i = i_t$. Then we have $h(d(x))_{i_0 \ldots i_p} = 0$. On the other hand we have

$$d(h(x))_{i_0 \ldots i_p} = \sum (-1)^j h(x)_{i_0 \ldots \hat{i}_j \ldots i_p} = (-1)^t h(x)_{i_0 \ldots \hat{i} \ldots i_p} = (-1)^t (-1)^{t-1} x_{i_0 \ldots i_p}$$

Thus $(dh + hd)(x)_{i_0 \ldots i_p} = -x_{i_0 \ldots i_p}$ as desired.

Case II: $i \notin \{i_0, \ldots, i_p\}$. Let j be such that $i_j < i < i_{j+1}$. Then we see that

$$\begin{aligned} h(d(x))_{i_0 \ldots i_p} &= (-1)^j d(x)_{i_0 \ldots i_j i i_{j+1} \ldots i_p} \\ &= \sum\nolimits_{j' \leq j} (-1)^{j+j'} x_{i_0 \ldots \hat{i}_{j'} \ldots i_j i i_{j+1} \ldots i_p} - x_{i_0 \ldots i_p} \\ &\quad + \sum\nolimits_{j' > j} (-1)^{j+j'+1} x_{i_0 \ldots i_j i i_{j+1} \ldots \hat{i}_{j'} \ldots i_p} \end{aligned}$$

On the other hand we have

$$\begin{aligned} d(h(x))_{i_0 \ldots i_p} &= \sum\nolimits_{j'} (-1)^{j'} h(x)_{i_0 \ldots \hat{i}_{j'} \ldots i_p} \\ &= \sum\nolimits_{j' \leq j} (-1)^{j'+j-1} x_{i_0 \ldots \hat{i}_{j'} \ldots i_j i i_{j+1} \ldots i_p} \\ &\quad + \sum\nolimits_{j' > j} (-1)^{j'+j} x_{i_0 \ldots i_j i i_{j+1} \ldots \hat{i}_{j'} \ldots i_p} \end{aligned}$$

Adding these up we obtain $(dh + hd)(x)_{i_0 \ldots i_p} = -x_{i_0 \ldots i_p}$ as desired. □